



LexTime: A Benchmark for Temporal Ordering of Legal Events

Bloomberg

Claire Barale¹, Leslie Barrett², Vikram Sunil Bajaj²,

Michael Rovatsos¹

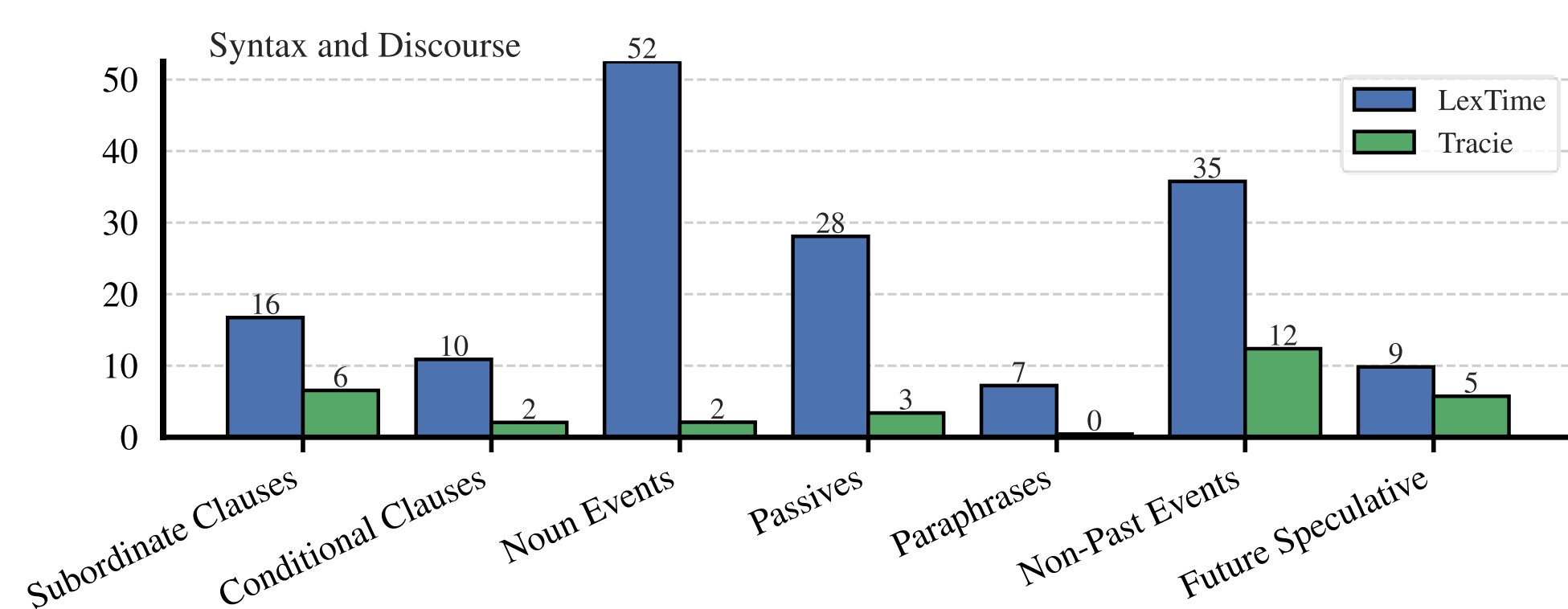
Engineering



Task overview

- Event Ordering → Reasoning about a sequence of events (QA task)
 - Event triggers: Verbs
 - Comparison with generic language, using a dataset of narrative texts (TRACIE, Zhou et al., 2021)
 - Dataset: Labor-related Federal complaints (2020-2024)
- 512 manually reviewed instances

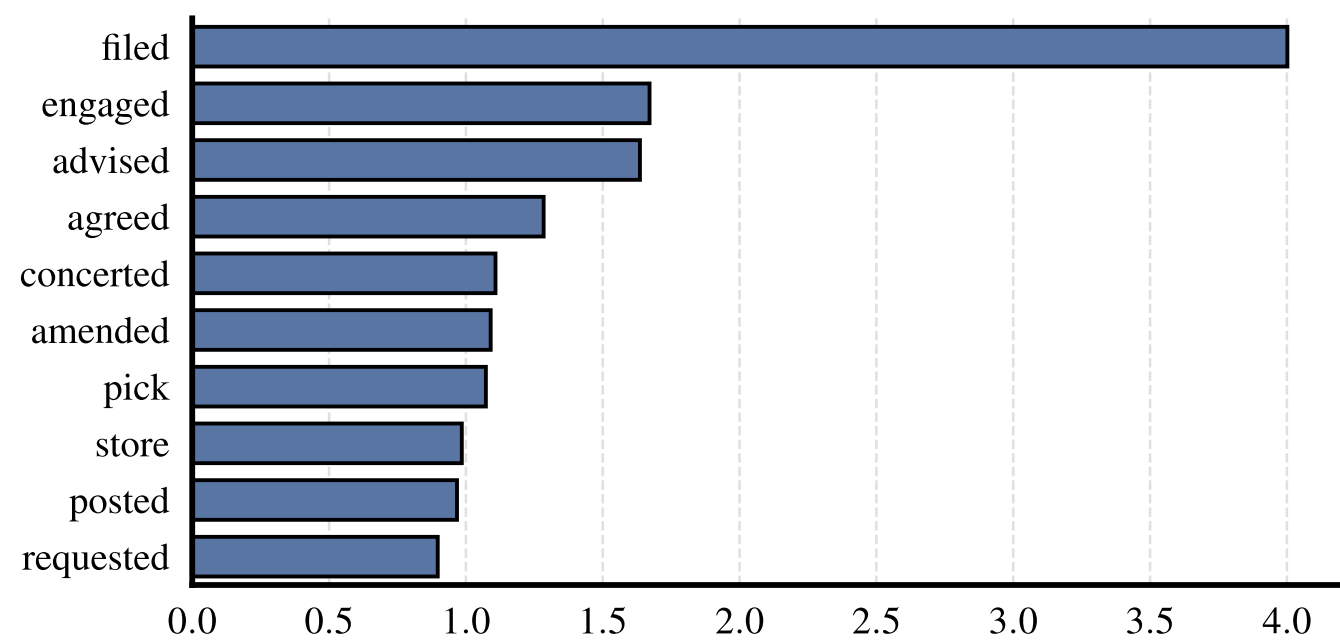
Linguistic features of events in legal texts



Comparison of linguistic features in LexTime and Tracie. Values (y-axis) represent the percentage of sentences that contain each feature, indicating its frequency in the text.

Lexical and syntactic differences (LexTime vs Tracie)

- Longer sentences (28 vs 9 tokens, median)
- Specialized legal vocabulary (*filed*, *granted*, *amended*)
- Higher event density (4.01 vs 1.60 events/sentence)
- More subordination (+10.5%) and modality (+11%)
- Frequent nominalization (+50.4%), paraphrasing (+6.56%), and passive voice (+24.76%)

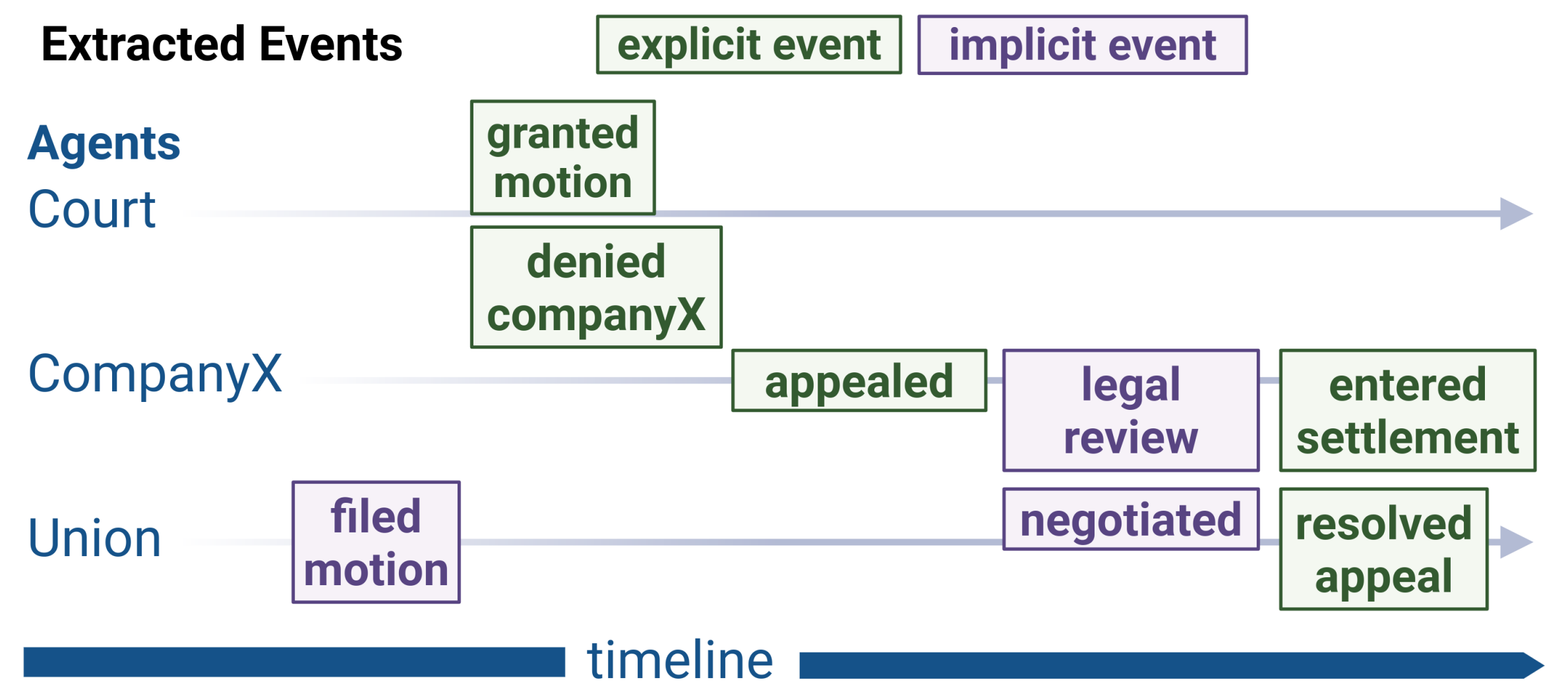


Top-10 triggers in LEXTIME by frequency (%)

Context paragraph

On November 25, 2020, the district court **granted** the unions summary judgment motion and **denied** CompanyX. CompanyX subsequently **appealed** the matter to the United States Court of Appeals for the Sixth Circuit (no. 20-xxxx). on September 17, 2021, however, the company and the union **entered** a settlement agreement that **resolved** the appeal.

Extracted Events



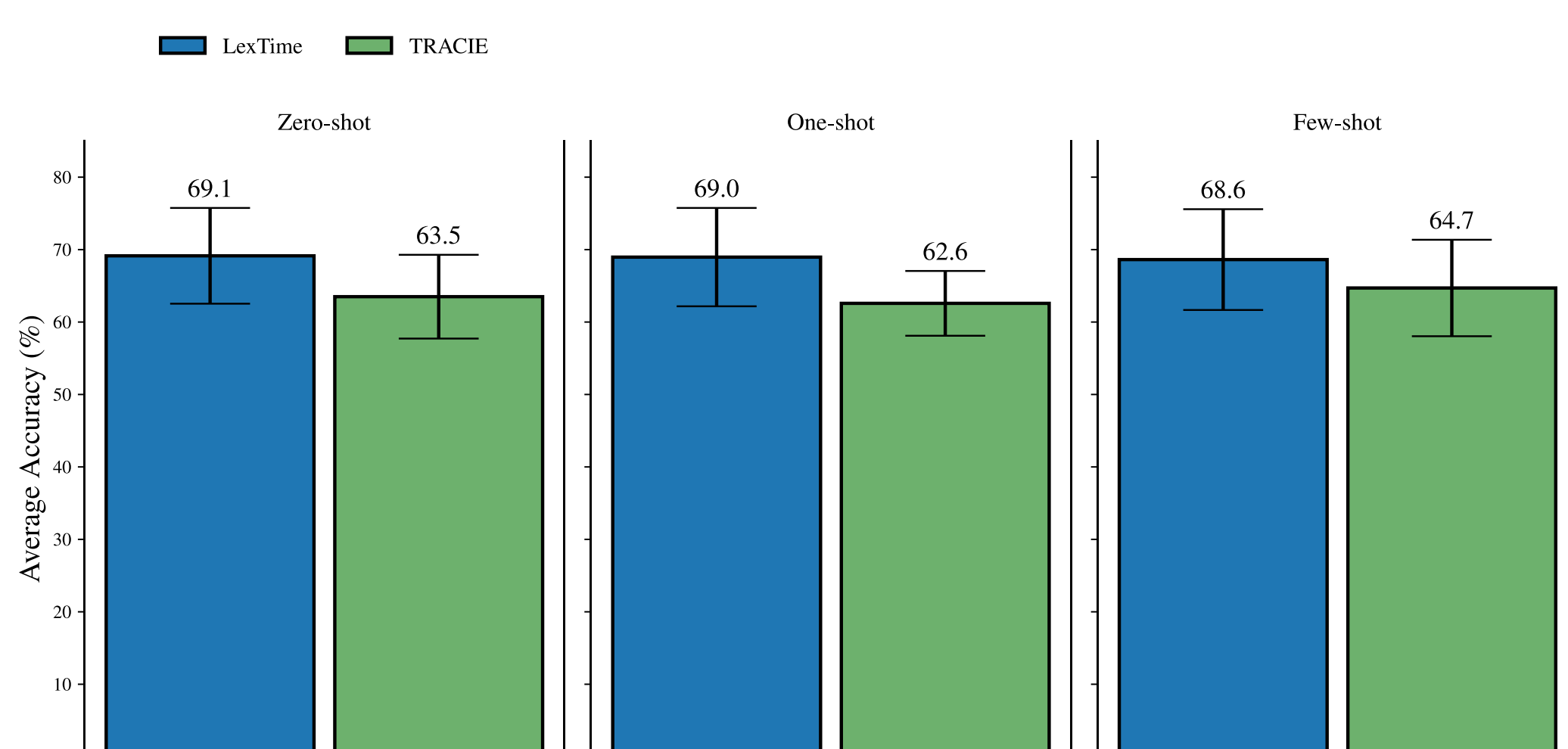
LexTime instance

Does the event **entering a settlement agreement** **follow** companyX **appeal**? **Yes**

(RQ) How do language models react to the distinct features of legal language for event ordering?

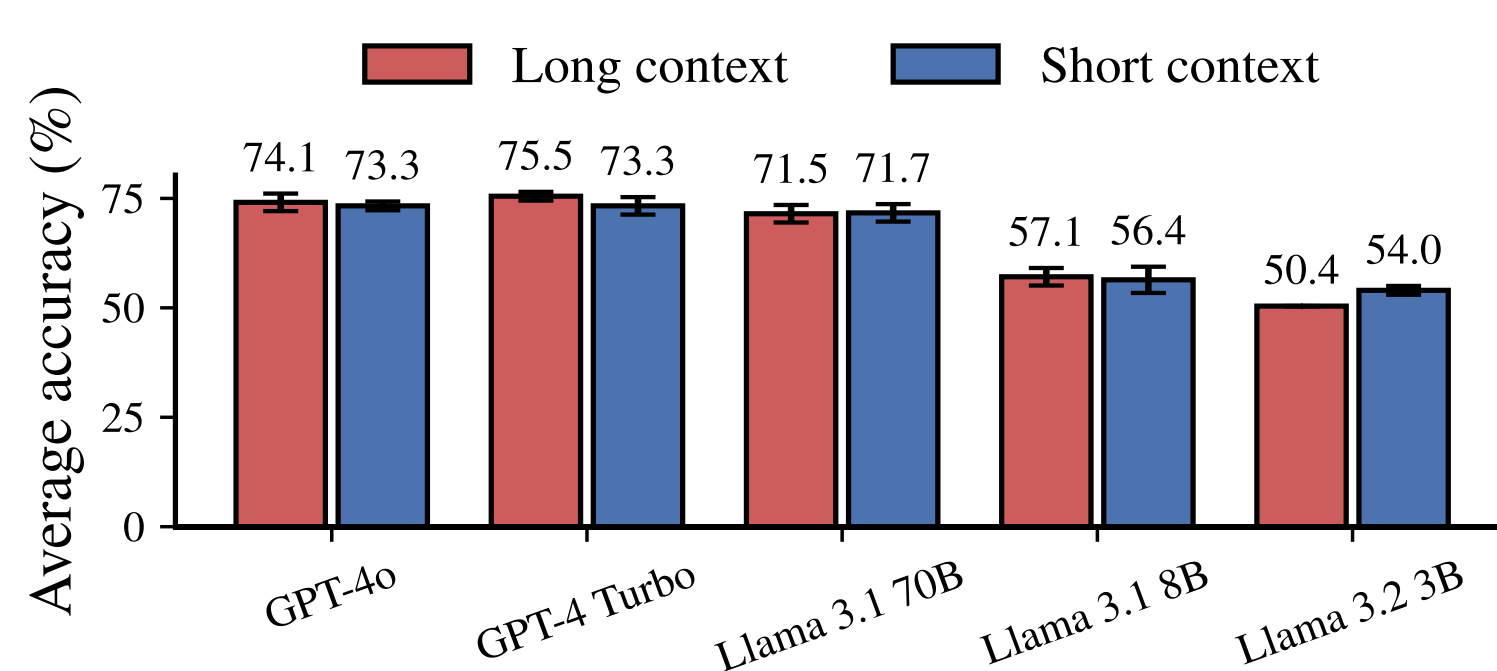
Results

Result 1: Models achieve higher event ordering accuracy on LexTime than on Tracie, by up to 10.5% (GPT-4o, few-shot)



Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B}, Llama 3.1_{8B} and Mistral_{123B}. 3 runs.

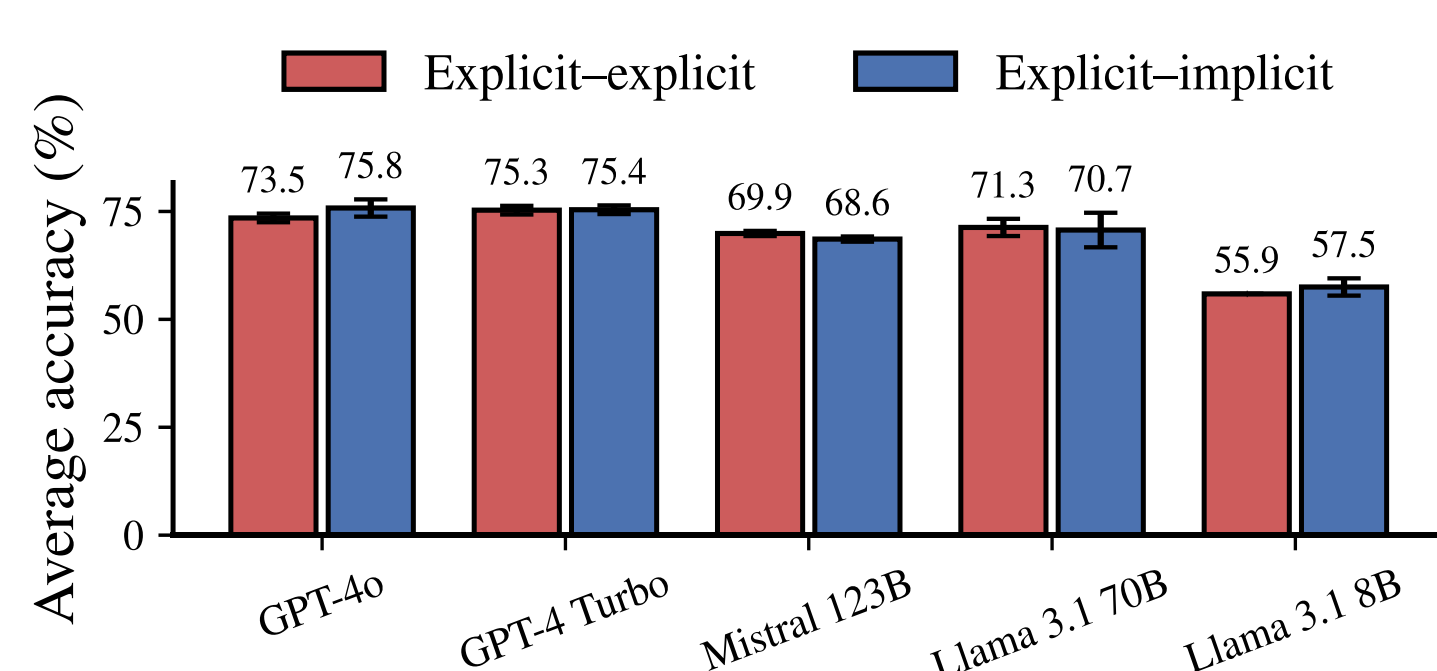
Result 2: Large models benefit from longer context windows



Context length:
- short <150 tokens
- long ≥150

Results are averaged from zero-shot, one-shot, few-shot settings. 3 runs.

Result 3: Large models perform better when implicit events are included

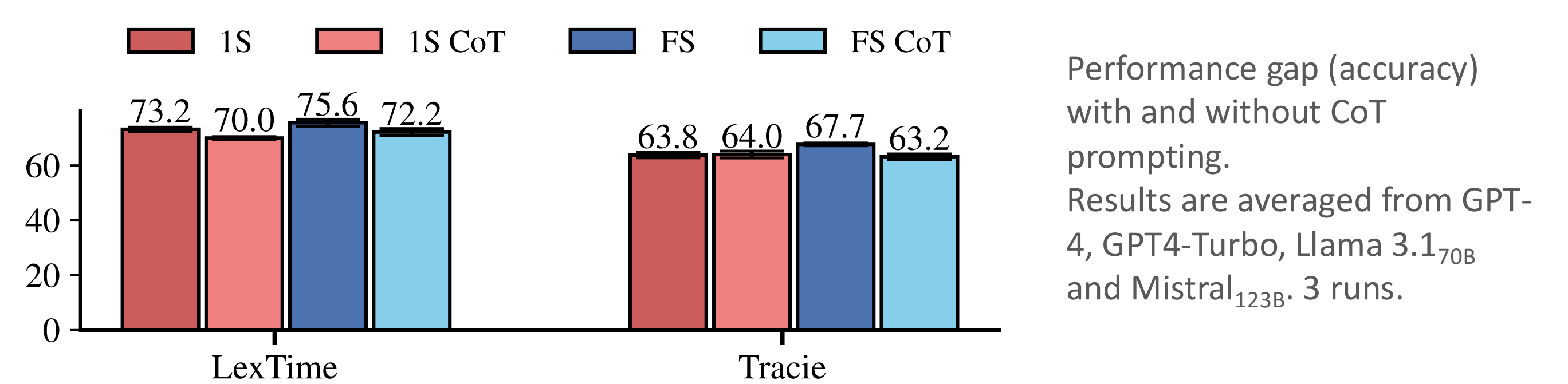


Explicit vs. implicit event pairs.

Query: Does *event A* happen before *event B*?
Example explicit-explicit pair: {*explicit event A*, *TR*, *explicit event B*}
Example explicit-implicit pair: {*explicit event A*, *TR*, *implicit event B*}

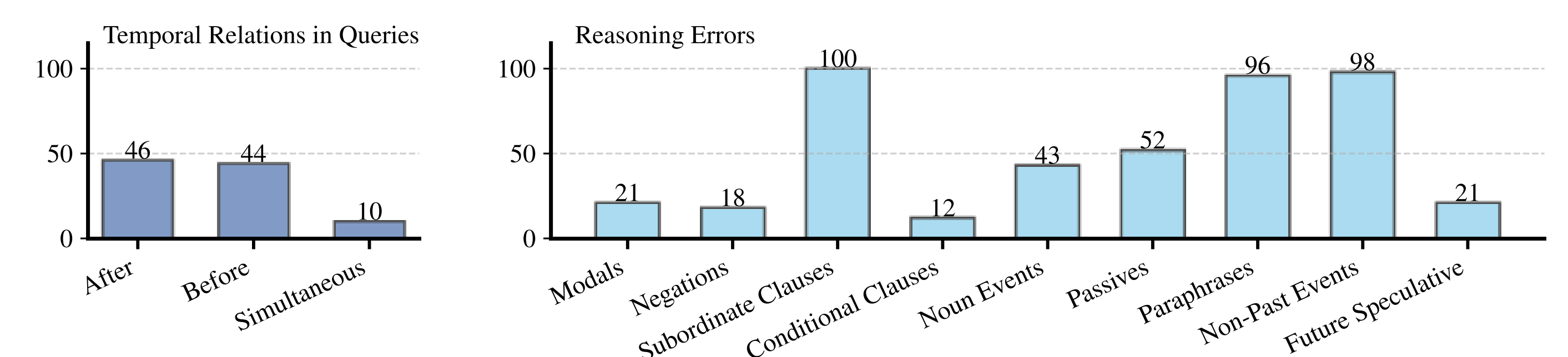
Results are averaged from zero-shot, one-shot, few-shot settings. 3 runs.

Result 4: Chain-of-thought prompting is not effective



Performance gap (accuracy) with and without CoT prompting. Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B} and Mistral_{123B}. 3 runs.

Result 5: Reasoning errors are associated with the linguistic specificities of legal language



Error analysis for LexTime on 100 misclassified samples from the few-shot setting across GPT-4, GPT-4 Turbo and Llama 3.1_{70B}.

Compared to the overall dataset, errors are associated with:

- +89% increase in paraphrases
- +84% in subordinate clauses
- +63% in events not in the past simple tense
- +24% in passive voice
- +12% in negations and future speculative constructions
- +6% in modality markers
- +2% in conditional clauses



© 2025 Bloomberg Finance L.P. All rights reserved.

The work was completed while Claire Barale was a Bloomberg Data Science Ph.D. Fellow.